PROGRAM:

```
/* program for binary search */
#include<stdio.h>
int BinarySearch(int arr[],int low ,int high,int key)
{
  int mid;
  while(low<=high)
  {
      mid=(low+high)/2;
      if (arr[mid]==key)
    return mid;
   if(arr[mid]<key)
  low=mid+1;
   else
  high=mid-1;
}
 return-1;
}
void print(int a[],int n )
{
int i;
for(i=0;i<n;i++) printf("%d ",a[i]);
}
int main()
{
      int arr[]={10,23,45,70,90,100,111,123};
      int key =45;
      int found_index;
      int n;
      n=sizeof(arr)/sizeof(int);
      print(arr,n);
      found_index=BinarySearch(arr,0,n-1,key);
      if(found_index!=-1)
            printf("\n key:%d search found at %d\n",key,found_index);
      else
      printf("\n key:%d search not found\n",key);
      key=150;
      found_index=BinarySearch(arr,0,n-1,key);
      if(found_index!=-1)
      printf("\n key:%d search found at %d\n",key,found_index);
      else
            printf("\n key:%d search not found\n",key);
      return 0;

}
```

OUTPUT:

```
10 23 45 70 90 100 111 123
key:45
search found at 2

key:150
search not found
```